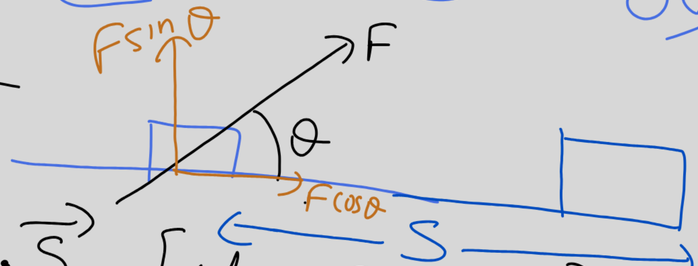
Ch-6
(Work & Energy)

Work (W)



$W = \vec{F} \cdot \vec{S}$ [when F = const] $= F S \cos\theta = (F\cos\theta)(S)$

$W = \int dw = \int \vec{F} \cdot d\vec{S}$ { when F = variable]

$W = F S \cos\theta$

N m = Joule [SI]

Dyne cm = erg [CGS]

$1\,J = ?\,erg$

$1\,J = 1\,N \times 1\,m$

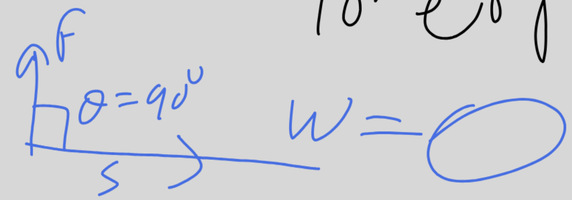
$= 10^5\,Dyne \times 10^2\,cm$

$= 10^7\,erg$

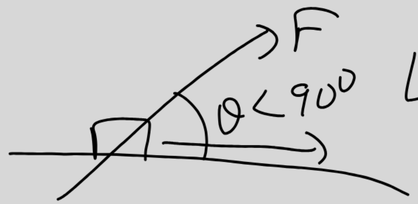
$W = F S \cos\theta \rightarrow +ve$

$0 \le \theta < 90$

$W = \quad 90 < \theta \le 180 \rightarrow -ve$



$W = 0$



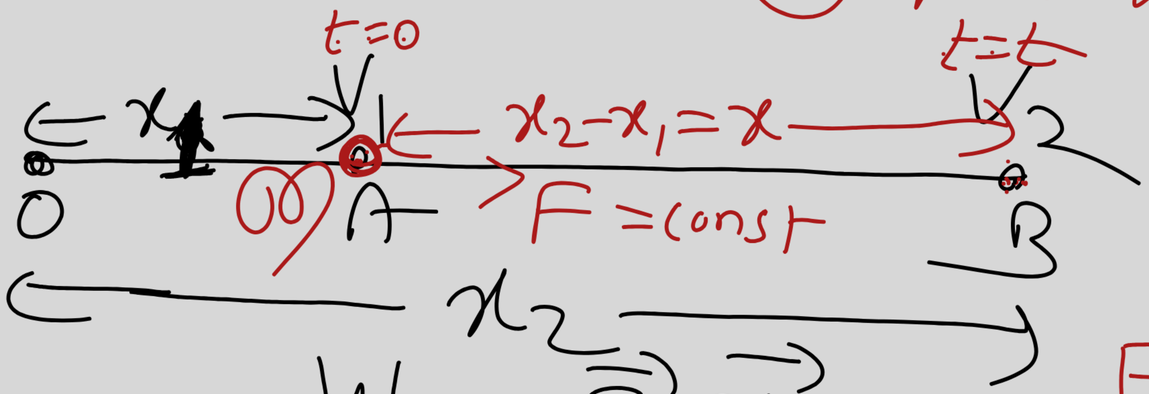
$W = +ve$

$W = F S \cos\theta$

$\cos 90 = 0$

Q Derive Work Energy Theorem (WET)

(i) For const force (ii) For variable force



$$W = \vec{F}\cdot\vec{x} = Fx\cos 0$$

$$v_2^2 - v_1^2 = 2a_x x$$

$$\Rightarrow a_x = \frac{v_2^2 - v_1^2}{2x}$$

$$F = m\left(\frac{v_2^2 - v_1^2}{2x}\right)$$

$$\Rightarrow W = \frac{m(v_2^2 - v_1^2)}{2x} \times x$$

$$\Rightarrow W = \frac{1}{2}m(v_2^2 - v_1^2) = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$$

$$= KE_f - KE_i$$

$$\boxed{W = \Delta KE}$$

WET for variable force: →

$$a_x = \frac{dv_x}{dt} = \frac{dv_x}{dx} \times \frac{dx}{dt} = v_x \frac{dv_x}{dx}$$

$$W = \int_{V_1}^{V_2} \vec{F} \cdot d\vec{x} = \int_{V_1}^{V_2} m a_x dx = \int_{V_1}^{V_2} m v_x \frac{dv_x}{dx} dx$$

$$\Rightarrow W = \int_{V_1}^{V_2} m v_x \, dv_x = m \frac{v_x^2}{2} \Big|_{V_1}^{V_2}$$

$$\Rightarrow W = \frac{1}{2} m (V_2^2 - V_1^2)$$

$$= \frac{1}{2} m V_2^2 - \frac{1}{2} m V_1^2$$

$$= KE_f - KE_i = \Delta KE = W$$

Power $P = \frac{dW}{dt} = \frac{\vec{F} \cdot (d\vec{s})}{dt} = \vec{F} \cdot \vec{v}$ = scalar

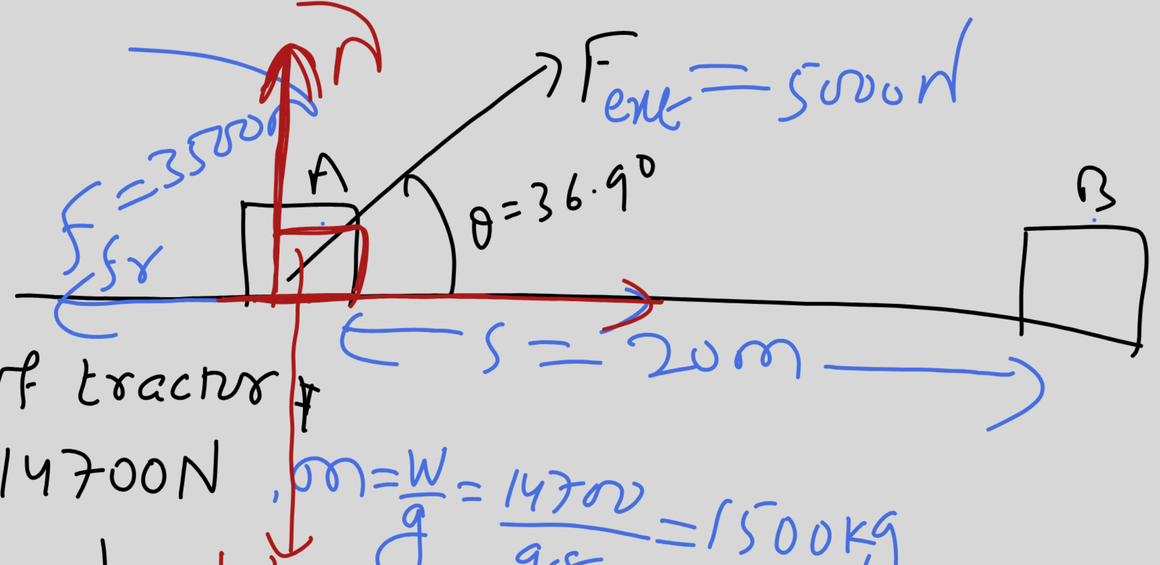
Unit of P. $\frac{\text{Joule}}{\text{sec}} = \text{Watt}$

$1\ hp = 746\ W = 0.746\ kW$

q8.9 p6

probability

Ex 6.2



Weight of tractor
$= 14700N$

$m = \frac{W}{g} = \frac{14700}{9.8} = 1500kg$

$W = mg$

$F_{ext} = 5000N$

$f_{frictional\ force} = 3500N$

$\theta = 36.9°$

$S = 20m$

Find Work Done by
(i) Grav. force
(ii) Normal reactⁿ
(iii) W_{ext} — 10KJ
(iv) W_{fric}
(v) W_{net}

$W_{grav} = 0$ = 

$W_{normal\ reactⁿ} = 0$

$W_{ext} = \vec{F}_{ext} \cdot \vec{S} = F_{ext} S \cos\theta = 5000 \times 20 \times \cos 36.9 = 10KJ$

$W_{fric} = \vec{F}_{fric} \cdot \vec{S} = F_{fric} S \cos 180 = -3500 \times 20 = -70KJ$

$W_{net} = 10KJ + (-70KJ)$

$= -60KJ$

mid sem soln (UPM oct'2026)

Q.1 (a) V_A A • $P' = \frac{1}{2}P_0$ $t_A = ?$; V_B B • $KE' = \frac{1}{2}KE$ $t_B = ?$; $\downarrow g$; $K_0 = \frac{1}{2}mu^2$

$P_0 = mu$ ↑ $u = 30$ m/s, O

(A) $P' = \frac{1}{2}P_0 \Rightarrow mV_A = \frac{1}{2}mu \Rightarrow V_A = \frac{u}{2} = \frac{30}{2} = 15$ m/s

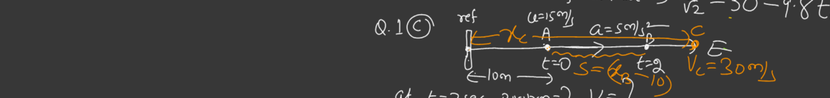
motion O to A: $u = 30$, $V = 15$, $a = -9.8$, $t = t_A = ?$

$V = u + at \Rightarrow 15 = 30 - 9.8\,t_A \Rightarrow t_A = 1.53$ sec

(B) $\frac{1}{2}mV_B^2 = \frac{1}{2}\left(\frac{1}{2}mu^2\right) \Rightarrow V_B = \frac{u}{\sqrt{2}} = \frac{30}{\sqrt{2}}$ m/s

$V_B = u + at \Rightarrow \frac{30}{\sqrt{2}} = 30 - 9.8\,t_B \Rightarrow t_B = 0.896$ sec

Q.1 (c)



at $t = 2$ sec. position = ? $V = ?$

x_B

A to B: $u = 15$ m/s, $a = 5$ m/s², $t = 2$ sec

$\Rightarrow S = ut + \frac{1}{2}at^2 \Rightarrow x_B - 10 = 15 \times 2 + \frac{1}{2}(5) \times 2^2$

$\Rightarrow x_B = 50$ m

$V = u + at = 15 + 5 \times 2 = 25$ m/s

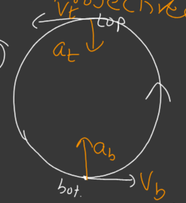
motion from A to C: $u = 15$, $V = 30$, $a = 5$, $S = ?$

$V^2 - u^2 = 2aS$

$\Rightarrow 30^2 - 15^2 = 2 \times 5 (x_c - 10) \Rightarrow x_c = 77.5$ m

Q.2 (a) P.t Projectile (Trajectory) is parabolic in nature

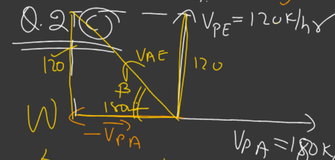
Q.2 (b)



$a_{bot} = 4\,a_{top}$

$\frac{V_{bot}^2}{R} = 4 \times \frac{V_{top}^2}{R} \Rightarrow V_{bot} = 2\,V_{top}$

Q.2 (c)



$\vec{V_{PE}}$, $\vec{V_{PA}}$, $\vec{V_{AE}} = ?$

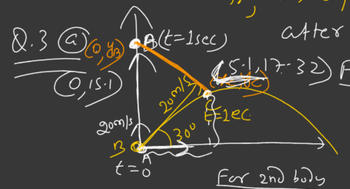
$\vec{V_{AE}} = \vec{V_{AP}} + \vec{V_{PE}} = (-\vec{V_{PA}}) + \vec{V_{PE}}$

$V_{AE} = \sqrt{180^2 + 120^2} = 216.33$ K/hr

$\tan\beta = \frac{120}{180} \Rightarrow \beta = 33.69^\circ$

Velocity of Air w.r.t Earth is $= 216.33$ K/hr, 33.69° N of W

Q.3 (a)



after $t = 1$ sec Find dist betn the two bodies

For 1st body (A to B): $u = 20$ m/s, $S = y_B$, $a = -9.8$ m/s², $t = 1$

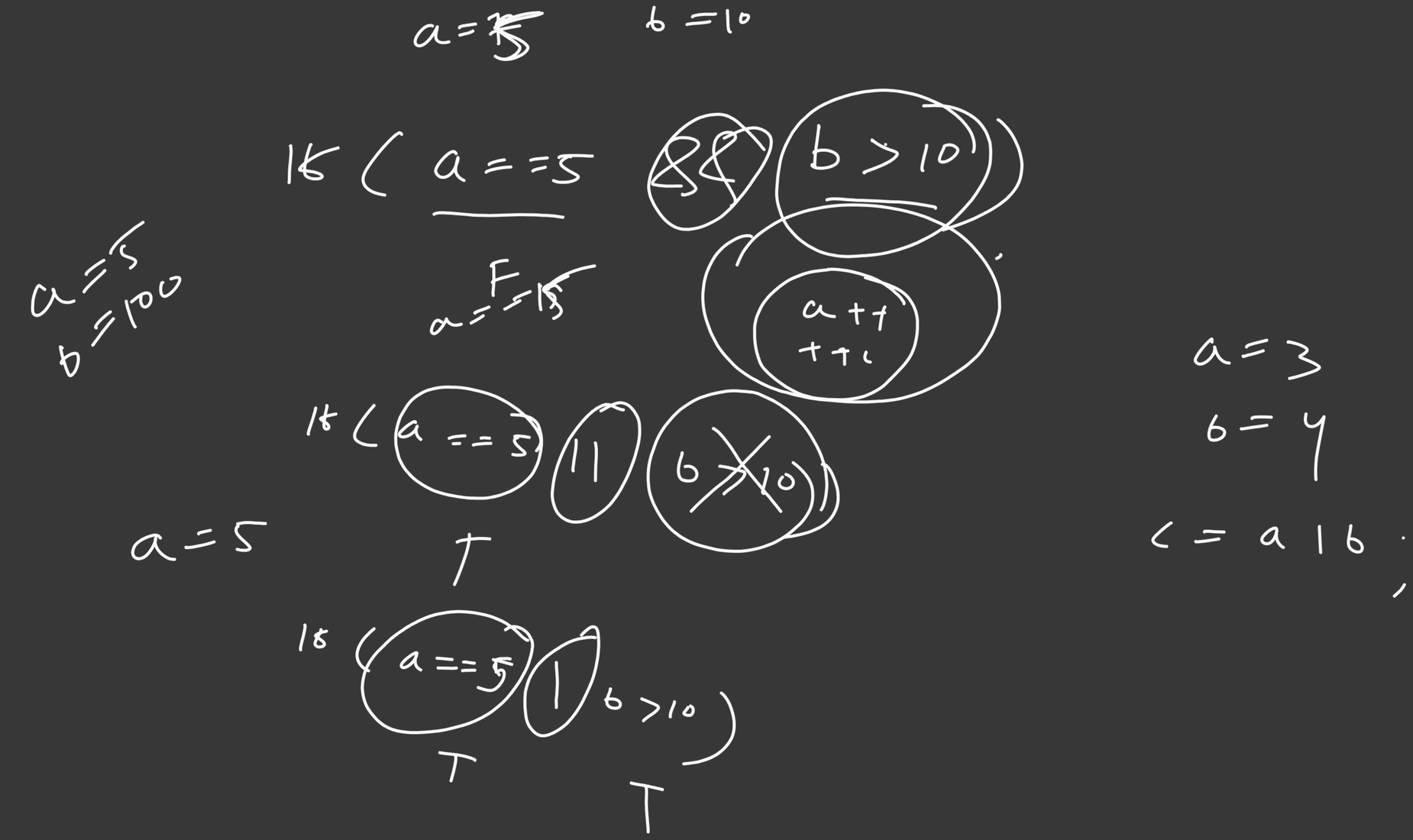
$S = ut + \frac{1}{2}at^2 \Rightarrow y_B = 20 \times 1 + \frac{1}{2}(-9.8) \times 1^2 = 15.1$ m

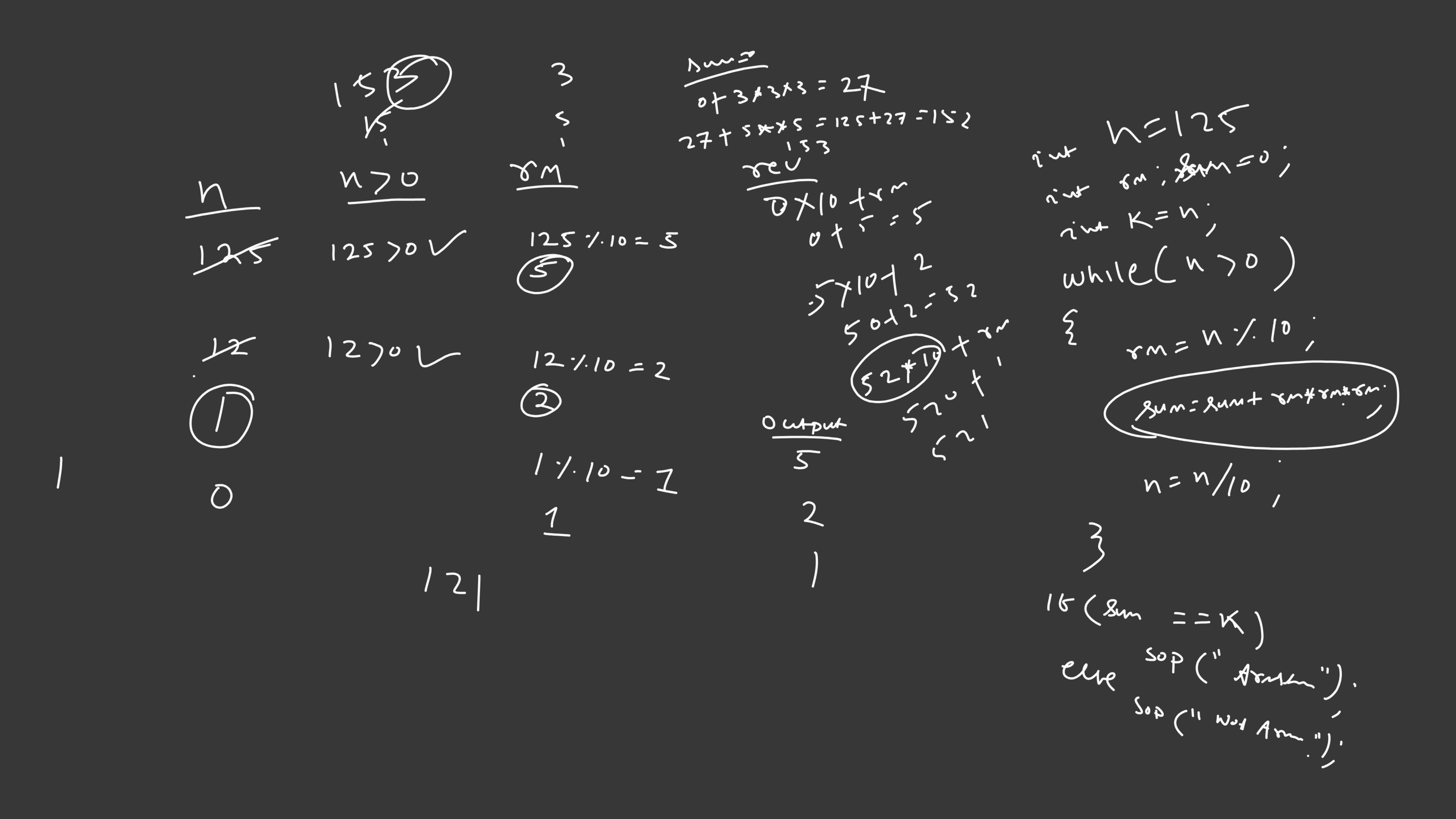
For 2nd body

$x = ut + \frac{1}{2}at^2 = 20\cos 30 \times 1 = 17.32$ m

$y = ut + \frac{1}{2}at^2 = 20\sin 30 \times 1 + \frac{1}{2}(-9.8) \times 1^2 = 5.1$ m

$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = 20$ m


a=5 b=10
if (a==5 && b>10)
a=5
b=100
a=F
a++
a=3
b=4
if (a == 5 || b>10)
a=5
T
if (a == 5 || b>10)
T
T
<= a || b ;


n
125
n>0
125>0 ✓
rm
125%10=5
5
12
12>0 ✓
12%10=2
2
1
1%10=1
1
0
121
Sum=
0+3*3*3=27
27+5*5*5=125+27=152
153
rev
0*10+rm
0+5=5
5*10+2
50+2=52
52*10+rm
520+1
521
output
5
2
1
int n=125
int rm; sum=0;
int K=n;
while(n>0)
{
rm=n%10;
sum=sum+rm*rm*rm;
n=n/10;
}
if(sum==K)
sop("Armstrong");
else
sop("Not Armstrong");

6

1, 2, 3, 6

n

10

1+2+3 = 6

```
int n=5;
int i=1;
int fact=1;
while(i<=n)
{
    fact=fact*i;
}
```

i	i<=n	n%i
1	1<=10 ✓	10%1==0 ✓
2	2<=10 ✓	10%2==0 ✓
3	3<=10 ✓	10%3==0 ✗
4	4<=10 ✓	10%4==0 ✗
5	5<=10 ✓	10%5==0 ✓
6	6<=10 ✓	10%6==4 ✗
.	.	.
10	10<=10 ✓	10%10==0

output

2
5
10

```
int n=10;
int c=0;
int i=1;
int sum=0;
while(i<=n)
{
    if(n%i==0)
        c=c+1;

    i++;
}

if(c==2)
    sop("prime");
else
    sop("Not prime");
```